

Buffalo Curve Analytics

How to Run & Test — Step-by-Step Guide

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1. What This Package Contains

Buffalo Curve Analytics is an offline web application that extracts the two curves (milk flow and vacuum pressure) from AMS milking-report PDFs and produces one CSV file per animal session. Everything runs locally — no internet connection is used at any point.

Item	Purpose
app.py · start.sh · frontend/	The web application
pipeline/pipeline.py	The computer-vision extraction pipeline
sample_input/sample_report.pdf	A real 2-page AMS report to test with (16 sessions)
demo/reports/	4 demo PDFs + exact ground truth (64 charts)
demo/score_demos.py	Scores extraction against that ground truth
pipeline/csv_synthetic_*/	Pre-generated pipeline output for the benchmark
Buffalo_Curve_Analytics_Explainer.pdf	Technical explainer (method & accuracy)
requirements.txt · Dockerfile	Dependencies / optional container setup

2. Requirements

Only **Python 3.10 or newer** is required (check with `python3 --version`). Tesseract OCR is **optional** — it is used only when processing PNG/JPG images instead of PDFs. All PDFs, including the included samples, work without it.

3. Setup (one time, ~2 minutes)

Open a terminal inside the `chart_digitizer` folder and run:

macOS / Linux

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

Windows (Command Prompt)

```
python -m venv venv
venv\Scripts\activate.bat
pip install -r requirements.txt
```

The install downloads OpenCV, NumPy, PyMuPDF, Pandas and Flask. It may take a minute or two on first run.

4. Run the Web App

macOS / Linux

```
./start.sh
```

Windows

```
python app.py
```

Then open `http://localhost:5001` in any browser and:

1. Drag `sample_input/sample_report.pdf` onto the upload area (or click it to browse).
2. Click **Run Extraction** — a progress bar shows charts being extracted.
3. **Expected result:** 16 sessions from 15 animals, dated 2024-03-07, with zero warnings.
4. Click any session row to preview the extracted curves and per-session statistics (peak flow, duration, estimated yield, average pressure).
5. Download individual CSVs, or all of them with **Download all as ZIP**.

Each CSV contains three columns — curve name, time in minutes (0–10), and value. Both curves start at (0, 0), and the flow curve runs to the end of the 10-minute session window.

```

curve,time_min,value
Flusso_kg_min,0.0,0.0
Flusso_kg_min,0.0202,0.1923
Pressione_assoluta_0.1bar,0.0,0.0
    
```

5. Command-Line Usage (optional)

The pipeline can also run without the web app:

```

python pipeline/pipeline.py --input sample_input/sample_report.pdf \
                             --out ./my_csv_output
    
```

Add `--report` to also write a JSON summary of every chart processed.

6. Verify the Accuracy Claims

The package includes three synthetic PDFs whose exact curve coordinates are known (`demo/reports/all_truth.json`), plus the pipeline's output for them. To reproduce the accuracy numbers reported in the technical explainer:

```

cd tests
python demo/score_demos.py
    
```

Expected output (48/48 charts matched):

Metric	Flusso (flow)	Pressione (pressure)
MAE — mean	0.12 kg/min	0.19 × 0.1 bar
MAE — median	0.11 kg/min	0.20 × 0.1 bar
Curve coverage	100.0 %	98.4 %

To regenerate the pipeline output from scratch before benchmarking (instead of using the pre-generated CSVs), run:

```

python pipeline/pipeline.py --input demo/reports/demo_1_normal.pdf \
                             --out pipeline/csv_synthetic_baseline
python pipeline/pipeline.py --input demo/reports/demo_2_mixed.pdf \
                             --out pipeline/csv_synthetic_multipage
python pipeline/pipeline.py --input demo/reports/demo_3_hard.pdf \
                             --out pipeline/csv_synthetic_edge
    
```

7. Docker (optional alternative)

If Docker is installed, Sections 2–3 can be skipped entirely:

```
docker build -t chart-digitizer .  
docker run -p 5001:5000 chart-digitizer
```

Then open `http://localhost:5001`.

8. Troubleshooting

Symptom	Fix
Port 5001 already in use	Run on another port: <code>./start.sh 5002</code> (then open <code>localhost:5002</code>)
ModuleNotFoundError	The virtual environment is not active — rerun the activate command from Section 3
Permission denied: <code>./start.sh</code>	<code>chmod +x start.sh</code>
'tesseract not found' warning	Only affects PNG/JPG inputs. PDFs work without Tesseract — no action needed

For anything else, contact the author.